

Mission 005 Coordinator Handoff

Verdict

****acceptable with caveats****

The repo now has the core Neon-backed architecture in place: a simple agent write endpoint, a Neon-backed store for reads/writes, preserved dashboard/latest-mission envelopes, and visible freshness/lag semantics. Backend, Frontend, Product, and QA handoffs have been refreshed, and the mission documentation PDFs now exist. The latest verification run confirmed local build/test/typecheck success and showed that production still serves Mission 004 / JSON-backed content, so the deployment/cutover gap remains open.

What is done

- Canonical trust boundary is implemented in code: backend validation + Neon DB are the source of truth.
- `POST /api/agent-events` exists as a simple server-validated write path into Neon, with payload-shape validation and DB-backed persistence.
- `GET /api/dashboard` and `GET /api/missions/latest` read from the Neon-backed store when DB env vars are present.
- The UI remains read-only and still consumes the same response envelopes.
- Freshness / lag / updated-at are preserved in the response contract.
- DB schema and migration scaffolding exist for `missions`, `mission\_events`, and write receipts.
- Mission 005 PDFs were generated and verified in `docs/missions/pdfs`.

Key gaps / caveats

1. ****Production cutover depends on deployment environment wiring.****
 - The runtime still falls back to JSON when `NEON\_DATABASE\_URL` / `DATABASE\_URL` is absent.
 - That fallback is acceptable for local/dev, but it must not remain the live truth path in production/preview after cutover.
2. ****Migration/backfill is not yet fully closed.****
 - There is a schema migration, but no clearly executed one-time JSON-to-Neon backfill path has been verified from this environment.
 - The boundary must be explicit: JSON may seed/bootstrap once, but new writes must go only to the simple backend API and Neon.
3. ****Live deployment still shows Mission 004 from this environment.****
 - The production URL is reachable, but it is not yet serving the Mission 005 cutover from the checkout in hand.
 - This is a deployment / environment wiring issue, not a local code-quality issue.
4. ****Freshness is mostly correct at the API layer, but DB persistence still stores a freshness field on events.****
 - Treat that as diagnostic/derived metadata, not a second source of truth.

- Read-time freshness should remain based on DB timestamps and response generation time.

Canonical DB trust boundary

- **Trusted:** Neon rows, backend validation, read API shaping.
- **Untrusted:** browser state, agent payloads before server validation, JSON files/feeds except as bootstrap/export artifacts.
- **Do not** treat JSON replication as the main architecture after migration.

Freshness semantics

- Freshness must remain visible as one of: `fresh`, `partial`, `delayed`, `stale`, `empty`.
- `updatedAt` and `lagMs` should be derived from DB state and response generation time.
- The UI should keep surfacing freshness/lag explicitly and remain read-only.

Migration / backfill boundary

- One-time historical import from the current JSON source is allowed.
- After backfill, writes must land only in the simple backend API and canonical DB.
- Any JSON export/debug artifact must be generated from DB state, not the reverse.
- Production/preview should not depend on JSON fallback once Neon is live.

Recommended next-step dependencies

Backend

- Finish any remaining deployment wiring checks for Neon-backed runtime in canonical environments.
- Confirm idempotency/replay behavior and sequence collision handling are still correct in live envs.
- Keep the write/read contract stable.

Frontend

- Keep the UI read-only.
- Continue consuming the existing dashboard/latest-mission envelopes.
- Ensure freshness / lag / updated-at remain visible and legible across states.

QA

- Verify writes are accepted with valid payloads and rejected for invalid payload shape.
- Verify accepted writes persist to Neon and appear through read APIs.
- Verify freshness/lag visibility in preview or production.
- Verify the app remains read-only for end users.
- Verify JSON is no longer the truth path after cutover.
- Re-run deployed smoke once the environment wiring is updated.

Coordinator call

Proceed with deployment/cutover verification next. Do not mark Mission 005 complete until the deployed app in the canonical environment is confirmed to be reading the Neon-backed path rather than the JSON fallback.